PHYS30762 Object Oriented Programming in C++

Assignment 5 Report

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Contents

1 Program summary **1**

2 References **2**

Note to markers: For version history please see the README file for different branches on the Github repository for assignment 5, as this contains planning and content for this report.

1. Program summary

- Program summary - What the code does (user guidelines?) - Inheritance diagrams - Code design - GAI

2. References

The following sources were used in the research for this assignment.

[1] https://www.geeksforgeeks.org/cpp/pure-virtual-functions-and-abstract-classes/

[2] https://en.cppreference.com/w/cpp/language/rule_of_three.html

[3] https://www.geeksforgeeks.org/cpp/understanding-constexper-specifier-in-cpp/

[4] https://www.geeksforgeeks.org/cpp/sqrt-sqrtl-sqrtf-cpp/

[5] https://www.geeksforgeeks.org/cpp/enum-classes-in-c-and-their-advantage-over-enum-datatype/

[6] https://www.geeksforgeeks.org/cpp/range-based-loop-c/

[7] https://en.wikipedia.org/wiki/Particle_decay

[8] https://physics.stackexchange.com/questions/282245/muon-decay-mode

[9] https://en.wikipedia.org/wiki/Muon

[10] https://en.wikipedia.org/wiki/Tau_(particle)

[11] https://tikz.net/sm_decay_matrix/

[12] https://www.geeksforgeeks.org/cpp/stduniform_real_-distribution-class-in-c-with-examples/

[13] https://www.w3schools.com/cpp/cpp_switch.asp

[14] https://www.geeksforgeeks.org/cpp/can-constructor-private-cpp/

[15] https://www.w3schools.com/cpp/cpp_constructors_overloading.asp

[16] https://www.geeksforgeeks.org/system-design/factory-method-pattern-c-design-patterns/

[17] https://pdg.lbl.gov/2020/AtomicNuclearProperties/HTML/tungsten_W.html